Chemistry Paper 3A WASSCE (SC), 2023

Question 2A

C is an Inorganic Compound.
D is an inorganic compound.
Carry out the following exercises on C and D.
Record your observations and identify any gas(es) evolved. State the conclusions you draw from the results of each test.

(a) Put C Into a test tube and add about 5 cm3 of distilled water and shake well. Divide the solution into two portions.
(i) Test the first portion with a litmus paper.
(ii) To the second portion, add about 1 cm of Fehling's solution, A and B and heat.

(b) Divide D into two portions.
(i) Put the first portion into a dry boiling tube and heat and then allow to cool.
(ii) Add about 5 *cm3* of dilute HCl to the second portion and heat, allow it to cool and filter necessary. Divide the resulting solution into two portions.
(ii) To the first portion from 2*(b)*(ii), add NaOH(aq) in drops and then in excess.
(iii) To the second portion from 2*(b)*(ii), add aqueous NH3 in drops and then in excess.

Observation

This question was attempted by majority of the candidates and their performance was above average.

In part (a), majority of the candidates tested the find portion of the solution with litmus paper, but some of them did not write the correct inference.

In part (b), majority of the candidates did not adhere to the instructions, and made them to lose marks.

The expected answers include:

C= Glucose D =ZnO

	TEST	OBSERVATION	INFERENCE
(a)	C + distilled water	Dissolved to form colourless solution	C is a soluble compound
(i)	C(aq) + litmus paper	No effect on litmus papers (red/blue)	Solution of C is neutral
(ii)	C(aq) + Fehling's solution + heat	A brick-red precipitate is formed	Reducing sugar/ reducing agent is present
b(i)	D + heat + cool	Solid turned yellow when hot, and turned white when cool	ZnO present
(ii)	D + HCl(aq) + heat	D dissolved to form a colourless solution	D is a base/basic oxide
(iii)	1st portion from b(ii) + NaOH(aq) in drops then in excess	A white gelatinous precipitate formed, Dissolves	Zn2+, Al3+ may be present Al3+, Zn2+ (both ions must be present to score)
(iv)	2nd portion from b(ii) + NH3(aq) in drops then in excess	White gelatinous precipitate formed precipitate dissolved	Zn2+, Al3+ Zn2+ is present

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